Theorem 5

The following are equivalent

(i) $A \in M_{n\times n}$ is diagonalizable

(ii) $A$ has $n$ linearly independent eigenvectors, $v_1, \dots, v_n \in \mathbb{R}^n$

(iii) $\mathbb{R}^n$ has a basis $\{v_1, \dots, v_n\}$ of eigenvectors of $A$.

(iv) $A = PDP^{-1}$ where

$P = [v_1 \cdots v_n]$ has a basis of eigenvectors of $A$

and $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$, where $Av_j = \lambda_j v_j$, $1 \le j \le n$.

Remarks

- Conceptually (iii) is the most important, as it is telling us that the linear transformation on $\mathbb{R}^n$, defined by multiplying with $A$, just multiplies basis by some scalars (the eigenvalues).

- Not all matrices are diagonalizable. If $A$ is a matrix coming from a mathematical model using data collected "from nature" then $A$ is almost certainly diagonalizable.

- $P = [v_1 \cdots v_n]$ in Thm 5 is actually a change-of-basis matrix. Let $\mathcal{B} = \{e_1, \dots, e_n\}$ (the standard basis) and $\mathcal{W} = \{v_1, \dots, v_n\}$

Then $v_1 = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = a_1 e_1 + \cdots + a_n e_n$

$$[v_1]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = v_1$$

Similarly $[v_j]_B = v_j$

By definition $\underset{B \leftarrow W}{P} = \left[ [v_1]_B \cdots [v_n]_B \right]$.

$$= [v_1 \cdots v_n] = P.$$

It follows that

$$A = PDP^{-1}$$

$$= \left(\underset{B \leftarrow W}{P}\right) D \left(\underset{B \leftarrow W}{P}\right)^{-1}$$

$$= \left(\underset{B \leftarrow W}{P}\right) D \left(\underset{W \leftarrow B}{P}\right)$$

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Theorem 6

Suppose $A \in M_{n \times n}$ has $n$ <u>distinct</u> eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. Then $A$ is diagonalizable.

Sketch of proof. By Thm 2 §5.1 the eigenvectors $v_1, \dots, v_n \in \mathbb{R}^n$ of $\lambda_1, \dots, \lambda_n$ form a lin. ind. set $\{v_1, \dots, v_n\}$. Since $\dim \mathbb{R}^n = n$ Thm 13 §4.5 tells us that $\{v_1, \dots, v_n\}$ is a basis of eigenvectors. Thm 5 (iii) tells us that $A$ is diagonalizable.

Example

Take $A = \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$

$$\det(A - xI) = \det\begin{bmatrix} 3-x & 1 \\ 0 & -1-x \end{bmatrix} = (3-x)(-1-x)$$

$\Rightarrow$ A two <u>distinct</u> eigenvalues $3, -1$.

Thm 6 $\Rightarrow$ A is diagonalizable.

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Example

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} I^{-1}$$

is diagonalizable, but there is only one eigenvalue.

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Example

Take $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$

$$\det\begin{bmatrix} 3-x & 1 \\ 0 & 3-x \end{bmatrix} = (3-x)^2$$

$\Rightarrow$ A has only 3 as an eigenvalue.

The 3-eigenspace is $\mathrm{Nul}(A - 3I) = \mathrm{Nul}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

$$= \left\{ \begin{bmatrix} c_1 \\ 0 \end{bmatrix} : c_2 \in \mathbb{R} \right\}$$

$$= \mathrm{span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

The 3-eigenspace has $\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}$ as basis and so is one-dimensional.

There are not enough lin. ind eigenvectors to give us a basis for $\mathbb{R}^2$. By Thm 5 (iii) $A$ is not diagonalizable.

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Thm 7 (Slightly different in textbook)

$A \in M_{n\times n}$ with exactly $p$ distinct eigenvalues. $\lambda_1, \dots, \lambda_p \in \mathbb{R}$.

(a) For any $1 \le k \le p$, the dimension of the $\lambda_k$-eigenspace ($\dim \mathrm{Nul}(A-\lambda_k I)$) is less than or equal to the multiplicity $m_k$ of $\lambda_k$ in $\det(A - xI)$.

(b) $A$ is diagonalizable $\iff$ (i) $\det(A-x) = (-1)^n (x-\lambda_1)^{m_1} \cdots (x-\lambda_p)^{m_p}$

(ii) $m_k = \dim \mathrm{Nul}(A-\lambda_k I)$

for <u>all</u> $1 \le k \le p$.

(c) Let $B_k$ be a basis for the $\lambda_k$-eigenspace ($\mathrm{Nul}(A-\lambda_k I)$).

Then $A$ is diagonalizable $\iff$ The sum of the dimensions of the eigenspaces is $n = \dim \mathbb{R}^n$.

$\iff B_1 \cup \dots \cup B_k$ is a basis of eigenvectors of $\mathbb{R}^n$.

Example

Take $A \in M_{4\times 4}$ with exactly 3 distinct eigenvalues, $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$. Suppose the dimension of the $\lambda_1$-eigenspace is 1, the dimension of the $\lambda_2$-eigenspace is 2. Is $A$ diagonalizable?

Let's look at the characteristic polynomial!

(*) $\det(A - xI) = (-1)^4 (x-\lambda_1)^{m_1} (x-\lambda_2)^{m_2} (x-\lambda_3)^{m_3} q(x)$

where $m_1, m_2, m_3$ are the multiplicities of $\lambda_1, \lambda_2$ and $\lambda_3$, and $q(x)$ is some polynomial.

Thm 7(a) tells us that

$$m_1 \geq \dim \mathrm{Nul}(A - \lambda_1 I) = 1$$
$$m_2 \geq \dim \mathrm{Nul}(A - \lambda_2 I) = 2$$
$$m_3 \geq \dim \mathrm{Nul}(A - \lambda_3 I) = ?$$

Since $\mathrm{Nul}(A - \lambda_3 I) = \lambda_3$-eigenspace

the $\lambda_3$-eigenspace contains eigenvectors for $\lambda_3$ and these eigenvectors are nonzero.

$\Rightarrow \mathrm{Nul}(A - \lambda_3 I) \neq \{0\}$

$\Rightarrow \dim(A - \lambda_3 I) \geq 1.$

$\Rightarrow m_3 \geq \dim(A - \lambda_3 I) \geq 1.$

Comparing the degree of $\det(A - xI) = 4$ in (*) with the right hand side we see that $4 = m_1 + m_2 + m_3 + \deg q(x) \geq m_1 + m_2 + m_3 \geq 1 + 2 + 1 = 4$

This forces $m_1 = 1$, $m_2 = 2$ and $m_3 = 1$ $(q(x) = 1)$
$(\deg q(x) = 0)$

By Thm 7(b) we see that $A$ is diagonalizable;

i.e. $m_1 = \dim \mathrm{Nul}(A - \lambda_1 I)$

$m_2 = \dim \mathrm{Nul}(A - \lambda_2 I)$

$m_3 = \dim \mathrm{Nul}(A - \lambda_1 I)$

and $\det(A - xI) = (x-\lambda_1)^1 (x-\lambda_2)^2 (x-\lambda_3)^1$.

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Example 3 § 5.3 in text.

Take $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.

$\det(A - xI) = -(x-1)^1 (x+2)^2$

$\Rightarrow \lambda_1 = 1$ and $\lambda_2 = -2$ are the eigenvalues of $A$

$\lambda_1$ has multiplicity 1, $\lambda_2$ has multiplicity 2.

Basis for the $\lambda_1$-eigenspace = basis for $\mathrm{Nul}(A - \lambda_1 I)$

$= \text{basis for span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

$= \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$.

$\Rightarrow$ dimension of $\lambda_1$-eigenspace is 1.

Basis for $\lambda_2$-eigenspace is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$\Rightarrow$ dimension of $\lambda_2$-eigenspace is 2.

Sum of dimensions of eigenspaces is $1+2=3=\dim \mathbb{R}^3$.

Also $m_1 = \dim \lambda_1$-eigenspace

$m_2 = \dim \lambda_2$-eigenspace

and $\det(A - xI) = (-1)^3 (x-\lambda_1)^1 (x-\lambda_2)^2$

By Thm 7 part (b) or (c) $A$ is diagonalizable

and a basis for $\mathbb{R}^3$ of eigenvectors is given by

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\} \cup \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

By Thm 6.

$$A = P \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} P^{-1}$$

where $P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$.